

QUI LOC PHAN

 Konatus

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 [locphan2001.github.io](https://github.com/locphan2001)

 Vietnamese

Education

Université Paris Dauphine & École des Mines de Paris

2023 – 2025

Master in Computer Science

Paris, France

- Graduated with highest grade in Combinatorial Optimization specialization of program
- Awarded Excellence Scholarship from partner institution ENS Paris Saclay and ENS Paris for 2 consecutive years

Ho Chi Minh University of Education

2019 – 2023

Honours Bachelor of Mathematics

Ho Chi Minh, Vietnam

- Graduated with Valedictorian Achievement (rank 1st/150)

Industrial Experience

Konatus, Startup of École Polytechnique

Sep 2025 – now

Research Engineer

Palaiseau, France

- Research on scheduling and resource allocation problems for large-scale corporate customers with complex projects
- Improve internal planning tools using evolutionary computation and machine learning to enhance decision efficiency
- Strengthen robustness and flexibility of solution by modeling uncertainty, adapting to resource variability and disruptions
- Evaluate and benchmark the company's solutions under different uncertainty models and operating conditions
- Collaborate with academic researchers to keep methods in line with cutting-edge techniques in operations research
- Work hand-in-hand with industry experts to deploy and validate methods on real historical and operational datasets

Corporate Research & Sciences Department, Dassault Systèmes

Mar 2025 – Sep 2025

Research Engineer Intern

Vélizy-Villacoublay, France

- Study quantum algorithms for constraint programming: optimization problems with integer and logical constraints
- Develop and implement for practical use cases in long-term, explore near-term challenges, conduct comparative analysis
- Practical case study on Healthcare Staff Scheduling at Paris Hospital with multi-objectives and multi-skills

Academic Experience

Optimization and Control Group, UMA, ENSTA Paris

May 2024 – Sep 2024

Graduate Research Intern

Palaiseau, France

- Study how to optimize resources used in quantum circuit with two objectives: number of gates and depth of circuit
- Understand and construct mathematical programming model and graph optimization model for the problems
- Design heuristics algorithms that produce good solutions and benchmark the results with available open-source libraries

HCMUE & Vietnam Academy of Science and Technology

10 months

Undergraduate Research Intern

Ho Chi Minh, Vietnam

- Study fundamental aspects of combinatorics and abstract algebra, model and address combinatorial problems
- Design or develop, and evaluate algebraic algorithms both in theory and practice mathematically

Project

Production Problem

2024

- Construct mathematical programming model for the resource allocation problem, inspired from the Flowshop Scheduling
- Implement Campbell - Dudek - Smith algorithm, with variant on preprocessing stage and postprocessing stage
- Perform clustering algorithms on preprocessing and local search methods on postprocessing to improve the solution

Quantum Optimization

2023

- Encode the mathematical programming model of Max-Cut and Maximum Independent Set into quantum computers
- Code end-to-end QAOA algorithms for the problem, test performance with different number of layers and parameters
- Benchmark with evolutionary algorithms on cost function and introduce some heuristic function for better solutions

Technical Skill

Language: Vietnamese (Native), English (C1), French (B1)

Programming Language: Python (3 years), C++ (2 years), Matlab (2 years)

Optimization Solvers: Gurobi, CPLEX, Google OR-Tools

Quantum Frameworks: Qiskit, PennyLane, Cirq